1) a)

$y = Ae^{-2x} + Be^{-3x}$

$\frac{dy}{dx} = -2Ae^{-2x} - 3Be^{-3x}$

$\frac{d^2y}{dx^2} = 4Ae^{-2x} + 9Be^{-3x}$

$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$

LHS: $4Ae^{-2x} + 9Be^{-3x} + 5(-2Ae^{-2x} - 3Be^{-3x}) + 6(Ae^{-2x} + Be^{-3x})$

$= 4Ae^{-2x} + 9Be^{-3x} - 10Ae^{-2x} - 15Be^{-3x} + 6Ae^{-2x} + 6Be^{-3x}$

$= (4 - 10 + 6)Ae^{-2x} + (9 - 15 + 6)Be^{-3x}$

$= 0$:RHS

1) b)

$y = k$

$\frac{dy}{dx} = 0$

$\frac{d^2y}{dx^2} = 0$

$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y$

$= 0 + 5 \times 0 + 6 \times k$

$= 6k$ as required

1) c)

$6K = 3 \Rightarrow K = \frac{1}{2}$

$\therefore y = \frac{1}{2}$ is a solution of the differential equation

1) d)

$y = A\mathrm{e}^{-2x} + B\mathrm{e}^{-3x} + \frac{1}{2}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2A\mathrm{e}^{-2x} - 3B\mathrm{e}^{-3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4A\mathrm{e}^{-2x} + 9B\mathrm{e}^{-3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 5\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$

LHS: $4A\mathrm{e}^{-2x} + 9B\mathrm{e}^{-3x} + 5(-2A\mathrm{e}^{-2x} - 3B\mathrm{e}^{-3x}) + 6\left(A\mathrm{e}^{-2x} + B\mathrm{e}^{-3x} + \frac{1}{2}\right)$

$= 4A\mathrm{e}^{-2x} + 9B\mathrm{e}^{-3x} - 10A\mathrm{e}^{-2x} - 15B\mathrm{e}^{-3x} + 6A\mathrm{e}^{-2x} + 6B\mathrm{e}^{-3x} + 6$

$= (4 - 10 + 6)A\mathrm{e}^{-2x} + (9 - 15 + 6)B\mathrm{e}^{-3x} + 6$

$= 0 + 0 + 6$

$= 6$:RHS

2) a)

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 20y = 5$

$\Rightarrow 0 - 0 - 20 \times k = 5$

$\Rightarrow k = -\frac{1}{4}$

Particular Integral:

$y = -\frac{1}{4}$

2) b)

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9y = 6$

$\Rightarrow 4 \times 0 - 9 \times k = 6$

$\Rightarrow k = -\frac{2}{3}$

Particular Integral:

$y = -\frac{2}{3}$

2) c)

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = -60$

$\Rightarrow 4 \times 0 + 11 \times 0 - 3 \times k = -60$

$\Rightarrow k = 20$

Particular Integral:

$y = 20$

3) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 20y = 5$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 20y = 0$

Auxiliary Equation:

$m^2 - m - 20 = 0$

$\Rightarrow (m-5)(m+4) = 0$

$\Rightarrow m = 5$ or $m = -4$

Complementary Function:

$y = A\mathrm{e}^{5x} + B\mathrm{e}^{-4x}$

Particular Integral:

$y = -\frac{1}{4}$

General Solution:

$y = A\mathrm{e}^{5x} + B\mathrm{e}^{-4x} - \frac{1}{4}$

3) b)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9y = 6$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9y = 0$

Auxiliary Equation:

$4m^2 - 9 = 0$

$\Rightarrow m^2 = \frac{9}{4}$

$\Rightarrow m = \pm\frac{3}{2}$

Complementary Function:

$y = A\mathrm{e}^{\frac{3}{2}x} + B\mathrm{e}^{-\frac{3}{2}x}$

Particular Integral:

$y = -\frac{2}{3}$

General Solution:

$y = A\mathrm{e}^{\frac{3}{2}x} + B\mathrm{e}^{-\frac{3}{2}x} - \frac{2}{3}$

3) c)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = -60$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} - 3y = 0$

Auxiliary Equation:

$4m^2 + 11m - 3 = 0$

$\Rightarrow (4m - 1)(m + 3) = 0$

$\Rightarrow m = \frac{1}{4}$ or $m = -3$

Complementary Function:

$y = A\mathrm{e}^{\frac{1}{4}x} + B\mathrm{e}^{-3x}$

Particular Integral:

$y = 20$

General Solution:

$y = A\mathrm{e}^{\frac{1}{4}x} + B\mathrm{e}^{-3x} + 20$

4) a)

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = 8$

$\Rightarrow 0 - 0 = 8$

As $0 \neq 8$, there is no value of k that satisfies the equation.

4) b)

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = 8$

$\Rightarrow 0 - a = 8$

$\Rightarrow a = -8$

The constant b can take on any value.

Particular Integral:

$y = -8x + b$

4) c)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = 8$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} = 0$

Auxiliary Equation:

$m^2 - m = 0$

$\Rightarrow m(m-1) = 0$

$\Rightarrow m = 0$ or $m = 1$

Complementary Function:

$y = A\mathrm{e}^0 + B\mathrm{e}^x$

$\Rightarrow y = A + B\mathrm{e}^x$

Particular Integral:

$y = -8x + b$

General Solution:

$y = A + B\mathrm{e}^x - 8x + b$

$\Rightarrow y = C + B\mathrm{e}^x - 8x$

where $C = A + b$

5)

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3\frac{\mathrm{d}y}{\mathrm{d}x} - 10y = 10x + 7$

$\Rightarrow 0 + 3a - 10(ax + b) = 10x + 7$

$\Rightarrow (-10a)x + (3a - 10b) = 10x + 7$

Compare coefficients:

$-10a = 10 \Rightarrow a = -1$

$3(-1) - 10b = 7 \Rightarrow -10b = 10 \Rightarrow b = -1$

Particular Integral:

$y = -x - 1$

$\therefore a = -1$ and $b = -1$

6) a)

$y = ax + b$

$\frac{dy}{dx} = a$

$\frac{d^2y}{dx^2} = 0$

$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

$\Rightarrow 0 + 7a - 30(ax + b) = 15x + 4$

$\Rightarrow (-30a)x + (7a - 30b) = 15x + 4$

Compare coefficients:

$-30a = 15 \Rightarrow a = -\frac{1}{2}$

$7\left(-\frac{1}{2}\right) - 30b = 4 \Rightarrow b = -\frac{1}{4}$

Particular Integral:

$y = -\frac{1}{2}x - \frac{1}{4}$

6) b)

$y = ax + b$

$\frac{dy}{dx} = a$

$\frac{d^2y}{dx^2} = 0$

$9\frac{d^2y}{dx^2} - y = 5 - x$

$\Rightarrow 0 - (ax + b) = 5 - x$

$\Rightarrow (-a)x - b = -x + 5$

Compare coefficients:

$-a = -1 \Rightarrow a = 1$

$-b = 5 \Rightarrow b = -5$

Particular Integral:

$y = x - 5$

6) c)

$y = ax + b$

$\frac{dy}{dx} = a$

$\frac{d^2y}{dx^2} = 0$

$4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$

$\Rightarrow 0 + 16a - 9(ax + b) = -45x$

$\Rightarrow (-9a)x + (16a - 9b) = -45x$

Compare coefficients:

$-9a = -45 \Rightarrow a = 5$

$16(5) - 9b = 0 \Rightarrow b = \frac{80}{9}$

Particular Integral:

$y = 5x + \frac{80}{9}$

7) a)

$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

Homogeneous Equation:

$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 0$

Auxiliary Equation:

$m^2 + 7m - 30 = 0$

$\Rightarrow (m + 10)(m - 3) = 0$

$\Rightarrow m = -10$ or $m = 3$

Complementary Function:

$y = Ae^{-10x} + Be^{3x}$

Particular Integral:

$y = -\frac{1}{2}x - \frac{1}{4}$

General Solution:

$y = Ae^{-10x} + Be^{3x} - \frac{1}{2}x - \frac{1}{4}$

7) b)

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - y = 5 - x$

Homogeneous Equation:

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - y = 0$

Auxiliary Equation:

$9m^2 - 1 = 0$

$\Rightarrow m^2 = \frac{1}{9}$

$\Rightarrow m = \pm\frac{1}{3}$

Complementary Function:

$y = A\mathrm{e}^{\frac{1}{3}x} + B\mathrm{e}^{-\frac{1}{3}x}$

Particular Integral:

$y = x - 5$

General Solution:

$y = A\mathrm{e}^{\frac{1}{3}x} + B\mathrm{e}^{-\frac{1}{3}x} + x - 5$

7) c)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 16\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = -45x$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 16\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = 0$

Auxiliary Equation:

$4m^2 + 16m - 9 = 0$

$\Rightarrow (2m + 9)(2m - 1) = 0$

$\Rightarrow m = -\frac{9}{2}$ or $m = \frac{1}{2}$

Complementary Function:

$y = A\mathrm{e}^{-\frac{9}{2}x} + B\mathrm{e}^{\frac{1}{2}x}$

Particular Integral:

$y = 5x + \frac{80}{9}$

General Solution:

$y = A\mathrm{e}^{-\frac{9}{2}x} + B\mathrm{e}^{\frac{1}{2}x} + 5x + \frac{80}{9}$

8) a)

$y = ax + b$

$\frac{dy}{dx} = a$

$\frac{d^2y}{dx^2} = 0$

$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$

$\Rightarrow 0 + 4a = 8x$

As the left hand side has no x but the right hand side does, there is no value of a that satisfies the equation.

8) b)

$y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{d^2y}{dx^2} = 2a$

$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$

$\Rightarrow 2a + 4(2ax + b) = 8x$

$\Rightarrow (8a)x + (2a + 4b) = 8x$

Compare Coefficients:

$8a = 8 \Rightarrow a = 1$

$2(1) + 4b = 0 \Rightarrow b = -\frac{1}{2}$

The constant c can take on any value.

Particular Integral:

$y = x^2 - \frac{1}{2}x + c$

8) c)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} = 8x$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

Auxiliary Equation:

$m^2 + 4m = 0$

$\Rightarrow m(m+4) = 0$

$\Rightarrow m = 0$ or $m = -4$

Complementary Function:

$y = A\mathrm{e}^0 + B\mathrm{e}^{-4x}$

$\Rightarrow y = A + B\mathrm{e}^{-4x}$

Particular Integral:

$y = x^2 - \frac{1}{2}x + c$

General Solution:

$y = A + B\mathrm{e}^{-4x} + x^2 - \frac{1}{2}x + c$

$\Rightarrow y = C + B\mathrm{e}^{-4x} + x^2 - \frac{1}{2}x$

where $C = A + c$

9)

$y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{d^2y}{dx^2} = 2a$

$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 12x^2 + 6x - 3$

$\Rightarrow 2a - (2ax + b) - 6(ax^2 + bx + c) = 12x^2 + 6x - 3$

$\Rightarrow (-6a)x^2 + (-2a - 6b)x + (2a - b - 6c) = 12x^2 + 6x - 3$

Compare coefficients:

$-6a = 12 \Rightarrow a = -2$

$-2(-2) - 6b = 6 \Rightarrow b = -\frac{1}{3}$

$2(-2) - \left(-\frac{1}{3}\right) - 6c = -3 \Rightarrow c = -\frac{1}{9}$

Particular Integral:

$y = -2x^2 - \frac{1}{3}x - \frac{1}{9}$

$\therefore a = -2$, $b = -\frac{1}{3}$ and $c = -\frac{1}{9}$

10) a)

$y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{d^2y}{dx^2} = 2a$

$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$

$\Rightarrow 2a - 8(2ax + b) - 9(ax^2 + bx + c) = 18x^2 + 11x - 14$

$\Rightarrow (-9a)x^2 + (-16a - 9b)x + (2a - 8b - 9c) = 18x^2 + 11x - 14$

Compare coefficients:

$-9a = 18 \Rightarrow a = -2$

$-16(-2) - 9b = 11 \Rightarrow b = \frac{7}{3}$

$2(-2) - 8\left(\frac{7}{3}\right) - 9c = -14 \Rightarrow c = -\frac{26}{27}$

Particular Integral:

$y = -2x^2 + \frac{7}{3}x - \frac{26}{27}$

10) b)

$y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{d^2y}{dx^2} = 2a$

$4\frac{d^2y}{dx^2} - 16y = -8x^2$

$\Rightarrow 4(2a) - 16(ax^2 + bx + c) = -8x^2$

$\Rightarrow (-16a)x^2 + (-16b)x + (8a - 16c) = -8x^2$

Compare coefficients:

$-16a = -8 \Rightarrow a = \frac{1}{2}$

$-16b = 0 \Rightarrow b = 0$

$8\left(\frac{1}{2}\right) - 16c = 0 \Rightarrow c = \frac{1}{4}$

Particular Integral:

$y = \frac{1}{2}x^2 + \frac{1}{4}$

10) c)

$y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{d^2y}{dx^2} = 2a$

$9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$

$\Rightarrow 9(2a) - 9(2ax + b) + 2(ax^2 + bx + c) = 11 - 4x - 10x^2$

$\Rightarrow (2a)x^2 + (-18a + 2b)x + (18a - 9b + 2c) = -10x^2 - 4x + 11$

Compare coefficients:

$2a = -10 \Rightarrow a = -5$

$-18(-5) + 2b = -4 \Rightarrow b = -47$

$18(-5) - 9(-47) + 2c = 11 \Rightarrow c = -161$

Particular Integral:

$y = -5x^2 - 47x - 161$

11) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 8\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = 18x^2 + 11x - 14$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 8\frac{\mathrm{d}y}{\mathrm{d}x} - 9y = 0$

Auxiliary Equation:

$m^2 - 8m - 9 = 0$

$\Rightarrow (m-9)(m+1) = 0$

$\Rightarrow m = 9$ or $m = -1$

Complementary Function:

$y = A\mathrm{e}^{9x} + B\mathrm{e}^{-x}$

Particular Integral:

$y = -2x^2 + \frac{7}{3}x - \frac{26}{27}$

General Solution:

$y = A\mathrm{e}^{9x} + B\mathrm{e}^{-x} - 2x^2 + \frac{7}{3}x - \frac{26}{27}$

11) b)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 16y = -8x^2$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 16y = 0$

Auxiliary Equation:

$4m^2 - 16 = 0$

$\Rightarrow m^2 - 4 = 0$

$\Rightarrow (m+2)(m-2) = 0$

$\Rightarrow m = -2$ or $m = 2$

Complementary Function:

$y = A\mathrm{e}^{-2x} + B\mathrm{e}^{2x}$

Particular Integral:

$y = \frac{1}{2}x^2 + \frac{1}{4}$

General Solution:

$y = A\mathrm{e}^{-2x} + B\mathrm{e}^{2x} + \frac{1}{2}x^2 + \frac{1}{4}$

11) c)

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 11 - 4x - 10x^2$

Homogeneous Equation:

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 9\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

Auxiliary Equation:

$9m^2 - 9m + 2 = 0$

$\Rightarrow (3m - 2)(3m - 1) = 0$

$\Rightarrow m = \frac{2}{3}$ or $m = \frac{1}{3}$

Complementary Function:

$y = A\mathrm{e}^{\frac{2}{3}x} + B\mathrm{e}^{\frac{1}{3}x}$

Particular Integral:

$y = -5x^2 - 47x - 161$

General Solution:

$y = A\mathrm{e}^{\frac{2}{3}x} + B\mathrm{e}^{\frac{1}{3}x} - 5x^2 - 47x - 161$

12) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 4x + 1$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$m^2 + 2m + 5 = 0$

$\Rightarrow (m+1)^2 - 1 + 5 = 0$

$\Rightarrow (m+1)^2 = -4$

$\Rightarrow m = -1 \pm 2\mathrm{i}$

Complementary Function:

$y = \mathrm{e}^{-x}(A\cos 2x + B\sin 2x)$

Try this Particular Integral:

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 4x + 1$

$\Rightarrow 0 + 2(a) + 5(ax + b) = 4x + 1$

$\Rightarrow (5a)x + (2a + 5b) = 4x + 1$

Compare coefficients:

$5a = 4 \Rightarrow a = \frac{4}{5}$

$2\left(\frac{4}{5}\right) + 5b = 1 \Rightarrow b = -\frac{3}{25}$

Particular Integral:

$y = \frac{4}{5}x - \frac{3}{25}$

General Solution:

$y = \mathrm{e}^{-x}(A\cos 2x + B\sin 2x) + \frac{4}{5}x - \frac{3}{25}$

12) b)

$\ddot{x} - 4\dot{x} + 4x = 4t^2 + 8t + 6$

Homogeneous Equation:

$\ddot{x} - 4\dot{x} + 4x = 0$

Auxiliary Equation:

$m^2 - 4m + 4 = 0$

$\Rightarrow (m-2)^2 = 0$

$\Rightarrow m = 2$

Complementary Function:

$x = Ate^{2t} + Be^{2t}$

Try this Particular Integral:

$x = at^2 + bt + c$

$\dot{x} = 2at + b$

$\ddot{x} = 2a$

$\ddot{x} - 4\dot{x} + 4x = 4t^2 + 8t + 6$

$\Rightarrow 2a - 4(2at + b) + 4(at^2 + bt + c) = 4t^2 + 8t + 6$

$\Rightarrow (4a)t^2 + (-8a + 4b)t + (2a - 4b + 4c) = 4t^2 + 8t + 6$

Compare coefficients:

$4a = 4 \Rightarrow a = 1$

$-8(1) + 4b = 8 \Rightarrow b = 4$

$2(1) - 4(4) + 4c = 6 \Rightarrow c = 5$

Particular Integral:

$x = t^2 + 4t + 5$

General Solution:

$x = Ate^{2t} + Be^{2t} + t^2 + 4t + 5$

12) c)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 9x$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3\mathrm{i}$

Complementary Function:

$y = A\cos 3x + B\sin 3x$

Try this Particular Integral:

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 9x$

$\Rightarrow 0 + 9(ax + b) = 9x$

$\Rightarrow (9a)x + (9b) = 9x$

Compare coefficients:

$9a = 9 \Rightarrow a = 1$

$9b = 0 \Rightarrow b = 0$

Particular Integral:

$y = x$

General Solution:

$y = A\cos 3x + B\sin 3x + x$

12) d)

$\ddot{x} - 5\dot{x} + 6x = 12$

Homogeneous Equation:

$\ddot{x} - 5\dot{x} + 6x = 0$

Auxiliary Equation:

$m^2 - 5m + 6 = 0$

$\Rightarrow (m-2)(m-3) = 0$

$\Rightarrow m = 2$ or $m = 3$

Complementary Function:

$x = Ae^{2t} + Be^{3t}$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$\ddot{x} - 5\dot{x} + 6x = 12$

$\Rightarrow 0 - 5(0) + 6(k) = 12$

$\Rightarrow k = 2$

Particular Integral:

$x = 2$

General Solution:

$x = Ae^{2t} + Be^{3t} + 2$

12) e)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 8x^2$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

Auxiliary Equation:

$m^2 + 4 = 0$

$\Rightarrow m^2 = -4$

$\Rightarrow m = \pm 2\mathrm{i}$

Complementary Function:

$y = A\cos 2x + B\sin 2x$

Try this Particular Integral:

$y = ax^2 + bx + c$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2ax + b$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 8x^2$

$\Rightarrow 2a + 4(ax^2 + bx + c) = 8x^2$

$\Rightarrow (4a)x^2 + (4b)x + (2a + 4c) = 8x^2$

Compare coefficients:

$4a = 8 \Rightarrow a = 2$

$4b = 0 \Rightarrow b = 0$

$2(2) + 4c = 0 \Rightarrow c = -1$

Particular Integral:

$y = 2x^2 - 1$

General Solution:

$y = A\cos 2x + B\sin 2x + 2x^2 - 1$

12) f)

$\ddot{x} + 4\dot{x} + 8x = 2t^2 + 4t$

Homogeneous Equation:

$\ddot{x} + 4\dot{x} + 8x = 0$

Auxiliary Equation:

$m^2 + 4m + 8 = 0$

$\Rightarrow (m+2)^2 - 4 + 8 = 0$

$\Rightarrow (m+2)^2 = -4$

$\Rightarrow m = -2 \pm 2\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-2t}(A\cos 2t + B\sin 2t)$

Try this Particular Integral:

$x = at^2 + bt + c$

$\dot{x} = 2at + b$

$\ddot{x} = 2a$

$\ddot{x} + 4\dot{x} + 8x = 2t^2 + 4t$

$\Rightarrow 2a + 4(2at + b) + 8(at^2 + bt + c) = 2t^2 + 4t$

$\Rightarrow (8a)t^2 + (8a + 8b)t + (2a + 4b + 8c) = 2t^2 + 4t$

Compare coefficients:

$8a = 2 \Rightarrow a = \frac{1}{4}$

$8\left(\frac{1}{4}\right) + 8b = 4 \Rightarrow b = \frac{1}{4}$

$2\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right) + 8c = 0 \Rightarrow c = -\frac{3}{16}$

Particular Integral:

$x = \frac{1}{4}t^2 + \frac{1}{4}t - \frac{3}{16}$

General Solution:

$x = \mathrm{e}^{-2t}(A\cos 2t + B\sin 2t) + \frac{1}{4}t^2 + \frac{1}{4}t - \frac{3}{16}$

12) g)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 4x + 2$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

Auxiliary Equation:

$m^2 - 3m + 2 = 0$

$\Rightarrow (m-1)(m-2) = 0$

$\Rightarrow m = 1$ or $m = 2$

Complementary Function:

$y = A\mathrm{e}^x + B\mathrm{e}^{2x}$

Try this Particular Integral:

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 4x + 2$

$\Rightarrow 0 - 3(a) + 2(ax + b) = 4x + 2$

$\Rightarrow (2a)x + (-3a + 2b) = 4x + 2$

Compare coefficients:

$2a = 4 \Rightarrow a = 2$

$-3(2) + 2b = 2 \Rightarrow b = 4$

Particular Integral:

$y = 2x + 4$

General Solution:

$y = A\mathrm{e}^x + B\mathrm{e}^{2x} + 2x + 4$

12) h)

$25\ddot{x} + 10\dot{x} + x = 6 - t^2$

Homogeneous Equation:

$25\ddot{x} + 10\dot{x} + x = 0$

Auxiliary Equation:

$25m^2 + 10m + 1 = 0$

$\Rightarrow (5m + 1)^2 = 0$

$\Rightarrow m = -\frac{1}{5}$

Complementary Function:

$x = Ate^{-\frac{1}{5}t} + Be^{-\frac{1}{5}t}$

Try this Particular Integral:

$x = at^2 + bt + c$

$\dot{x} = 2at + b$

$\ddot{x} = 2a$

$25\ddot{x} + 10\dot{x} + x = 6 - t^2$

$\Rightarrow 25(2a) + 10(2at + b) + (at^2 + bx + c) = 6 - t^2$

$\Rightarrow (a)t^2 + (20a + b)t + (50a + 10b + c) = -t^2 + 6$

Compare coefficients:

$a = -1$

$20(-1) + b = 0 \Rightarrow b = 20$

$50(-1) + 10(20) + c = 6 \Rightarrow c = -144$

Particular Integral:

$x = -t^2 + 20t - 144$

General Solution:

$x = Ate^{-\frac{1}{5}t} + Be^{-\frac{1}{5}t} - t^2 + 20t - 144$

13)

$\frac{d^2y}{dx^2} + 25y = 50x^2 + 25x$

Homogeneous Equation:

$\frac{d^2y}{dx^2} + 25y = 0$

Auxiliary Equation:

$m^2 + 25 = 0$

$\Rightarrow m^2 = -25$

$\Rightarrow m = \pm 5\mathrm{i}$

Complementary Function:

$y = A\cos 5x + B\sin 5x$

Try this Particular Integral:

$y = ax^2 + bx + c$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2ax + b$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 25y = 50x^2 + 25x$

$\Rightarrow 2a + 25(ax^2 + bx + c) = 50x^2 + 25x$

$\Rightarrow (25a)x^2 + (25b)x + (2a + 25c) = 50x^2 + 25x$

Compare coefficients:

$25a = 50 \Rightarrow a = 2$

$25b = 25 \Rightarrow b = 1$

$2(2) + 25c = 0 \Rightarrow c = -\frac{4}{25}$

Particular Integral:

$y = 2x^2 + x - \frac{4}{25}$

General Solution:

$y = A\cos 5x + B\sin 5x + 2x^2 + x - \frac{4}{25}$

Substitute $y = 0$ and $x = 0$:

$0 = A\cos 0 + B\sin 0 + 0 + 0 - \frac{4}{25}$

$\Rightarrow 0 = A - \frac{4}{25} \Rightarrow A = \frac{4}{25}$

$\therefore y = \frac{4}{25}\cos 5x + B\sin 5x + 2x^2 + x - \frac{4}{25}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{4}{5}\sin 5x + 5B\cos 5x + 4x + 1$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$ and $x = 0$:

$1 = -\frac{4}{5}\sin 0 + 5B\cos 0 + 0 + 1$

$\Rightarrow 1 = 5B + 1 \Rightarrow B = 0$

Particular Solution:

$y = \frac{4}{25}\cos 5x + 2x^2 + x - \frac{4}{25}$

14)

$4\ddot{x} - 4\dot{x} + x = 4t + 5$

Homogeneous Equation:

$4\ddot{x} - 4\dot{x} + x = 0$

Auxiliary Equation:

$4m^2 - 4m + 1 = 0$

$\Rightarrow (2m - 1)^2 = 0$

$\Rightarrow m = \frac{1}{2}$

Complementary Function:

$x = Ate^{\frac{1}{2}t} + Be^{\frac{1}{2}t}$

Try this Particular Integral:

$x = at + b$

$\dot{x} = a$

$\ddot{x} = 0$

$4\ddot{x} - 4\dot{x} + x = 4t + 5$

$\Rightarrow 0 - 4(a) + (at + b) = 4t + 5$

$\Rightarrow (a)t + (-4a + b) = 4t + 5$

Compare coefficients:

$a = 4$

$-4(4) + b = 5 \Rightarrow b = 21$

Particular Integral:

$x = 4t + 21$

General Solution:

$x = Ate^{\frac{1}{2}t} + Be^{\frac{1}{2}t} + 4t + 21$

Substitute $x = 1$ and $t = 0$:

$1 = A \times 0 \times e^0 + Be^0 + 4 \times 0 + 21$

$\Rightarrow 1 = B + 21 \Rightarrow B = -20$

$\therefore x = Ate^{\frac{1}{2}t} - 20e^{\frac{1}{2}t} + 4t + 21$

$\dot{x} = \frac{1}{2}Ate^{\frac{1}{2}t} + Ae^{\frac{1}{2}t} - 10e^{\frac{1}{2}t} + 4$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \frac{1}{2}A \times 0 \times e^0 + Ae^0 - 10e^0 + 4$

$\Rightarrow 0 = A - 10 + 4 \Rightarrow A = 6$

Particular Solution: $x = 6te^{\frac{1}{2}t} - 20e^{\frac{1}{2}t} + 4t + 21$

15)

$y = ke^x$

$\frac{dy}{dx} = ke^x$

$\frac{d^2y}{dx^2} = ke^x$

$\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 12e^x$

$\Rightarrow ke^x - 9(ke^x) + 14(ke^x) = 12e^x$

$\Rightarrow (1 - 9 + 14)ke^x = 12e^x$

$\Rightarrow 6k = 12$

$\Rightarrow k = 2$

Particular Integral:

$y = 2e^x$

$\therefore k = 2$

16)

$y = ke^{2x}$

$\frac{dy}{dx} = 2ke^{2x}$

$\frac{d^2y}{dx^2} = 4ke^{2x}$

$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 8e^{2x}$

$\Rightarrow 4ke^{2x} - (2ke^{2x}) - 12(ke^{2x}) = 8e^{2x}$

$\Rightarrow (4 - 2 - 12)ke^{2x} = 8e^{2x}$

$\Rightarrow -10k = 8$

$\Rightarrow k = -\frac{4}{5}$

Particular Integral:

$y = -\frac{4}{5}e^{2x}$

$\therefore k = -\frac{4}{5}$

17) a)

$y = ke^{4x}$

$\frac{dy}{dx} = 4ke^{4x}$

$\frac{d^2y}{dx^2} = 16ke^{4x}$

$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

$\Rightarrow 16ke^{4x} - 3(4ke^{4x}) + 2(ke^{4x}) = 5e^{4x}$

$\Rightarrow (16 - 12 + 2)ke^{4x} = 5e^{4x}$

$\Rightarrow 6k = 5$

$\Rightarrow k = \frac{5}{6}$

Particular Integral:

$y = \frac{5}{6}e^{4x}$

17) b)

$y = ke^{-2x}$

$\frac{dy}{dx} = -2ke^{-2x}$

$\frac{d^2y}{dx^2} = 4ke^{-2x}$

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

$\Rightarrow 4ke^{-2x} + (-2ke^{-2x}) - 6(ke^{-2x}) = 3e^{-2x}$

$\Rightarrow (4 - 2 - 6)ke^{-2x} = 3e^{-2x}$

$\Rightarrow -4k = 3$

$\Rightarrow k = -\frac{3}{4}$

Particular Integral:

$y = -\frac{3}{4}e^{-2x}$

17) c)

$y = ke^x$

$\frac{dy}{dx} = ke^x$

$\frac{d^2y}{dx^2} = ke^x$

$2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^x$

$\Rightarrow 2(ke^x) - 5(ke^x) - 3(ke^x) = 4e^x$

$\Rightarrow (2 - 5 - 3)ke^x = 4e^x$

$\Rightarrow -6k = 4$

$\Rightarrow k = -\frac{2}{3}$

Particular Integral:

$y = -\frac{2}{3}e^x$

18) a)

$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

Homogeneous Equation:

$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

Auxiliary Equation:

$m^2 - 3m + 2 = 0$

$\Rightarrow (m - 1)(m - 2) = 0$

$\Rightarrow m = 1$ or $m = 2$

Complementary Function:

$y = Ae^x + Be^{2x}$

Particular Integral:

$y = \frac{5}{6}e^{4x}$

General Solution:

$y = Ae^x + Be^{2x} + \frac{5}{6}e^{4x}$

18) b)

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

Homogeneous Equation:

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$

Auxiliary Equation:

$m^2 + m - 6 = 0$

$\Rightarrow (m+3)(m-2) = 0$

$\Rightarrow m = -3$ or $m = 2$

Complementary Function:

$y = Ae^{-3x} + Be^{2x}$

Particular Integral:

$y = -\frac{3}{4}e^{-2x}$

General Solution:

$y = Ae^{-3x} + Be^{2x} - \frac{3}{4}e^{-2x}$

18) c)

$2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$

Homogeneous Equation:

$2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 0$

Auxiliary Equation:

$2m^2 - 5m - 3 = 0$

$\Rightarrow (2m+1)(m-3) = 0$

$\Rightarrow m = -\frac{1}{2}$ or $m = 3$

Complementary Function:

$y = Ae^{-\frac{1}{2}x} + Be^{3x}$

Particular Integral:

$y = -\frac{2}{3}e^{x}$

General Solution:

$y = Ae^{-\frac{1}{2}x} + Be^{3x} - \frac{2}{3}e^{x}$

19) a)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 8\frac{\mathrm{d}y}{\mathrm{d}x} + 15y = 0$

Auxiliary Equation:

$m^2 - 8m + 15 = 0$

$\Rightarrow (m-3)(m-5) = 0$

$\Rightarrow m = 3$ or $m = 5$

Complementary Function:

$y = A\mathrm{e}^{3x} + B\mathrm{e}^{5x}$

19) b)

$y = k\mathrm{e}^{3x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3k\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 9k\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 8\frac{\mathrm{d}y}{\mathrm{d}x} + 15y = 6\mathrm{e}^{3x}$

$\Rightarrow 9k\mathrm{e}^{3x} - 8(3k\mathrm{e}^{3x}) + 15(k\mathrm{e}^{3x}) = 6\mathrm{e}^{3x}$

$\Rightarrow (9 - 24 + 15)k\mathrm{e}^{3x} = 6\mathrm{e}^{3x}$

$\Rightarrow 0 = 6$

There is no value of k for which this is true.

19) c)

$y = kxe^{3x}$

$\frac{dy}{dx} = 3kxe^{3x} + ke^{3x}$

$\frac{d^2y}{dx^2} = 9kxe^{3x} + 3ke^{3x} + 3ke^{3x}$

$= 9kxe^{3x} + 6ke^{3x}$

$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 6e^{3x}$

$\Rightarrow (9kxe^{3x} + 6ke^{3x}) - 8(3kxe^{3x} + ke^{3x}) + 15(kxe^{3x}) = 6e^{3x}$

$\Rightarrow (9 - 24 + 15)kxe^{3x} + (6 - 8)ke^{3x} = 6e^{3x}$

$\Rightarrow 0 - 2ke^{3x} = 6e^{3x}$

$\Rightarrow -2k = 6$

$\Rightarrow k = -3$

Particular Integral:

$y = -3xe^{3x}$

$\therefore k = -3$

19) d)

General Solution:

$y = Ae^{3x} + Be^{5x} - 3xe^{3x}$

20) a)

$2\ddot{x} - 5\dot{x} - 3x = 6e^{3t}$

Homogeneous Equation:

$2\ddot{x} - 5\dot{x} - 3x = 0$

Auxiliary Equation:

$2m^2 - 5m - 3 = 0$

$\Rightarrow (2m + 1)(m - 3) = 0$

$\Rightarrow m = -\frac{1}{2}$ or $m = 3$

Complementary Function:

$x = Ae^{-\frac{1}{2}t} + Be^{3t}$

Try this Particular Integral:

$x = kte^{3t}$

$\dot{x} = 3kte^{3t} + ke^{3t}$

$\ddot{x} = 9kte^{3t} + 3ke^{3t} + 3ke^{3t}$

$= 9kte^{3t} + 6ke^{3t}$

$2\ddot{x} - 5\dot{x} - 3x = 6e^{3t}$

$\Rightarrow 2(9kte^{3t} + 6ke^{3t}) - 5(3kte^{3t} + ke^{3t}) - 3(kte^{3t}) = 6e^{3t}$

$\Rightarrow (18 - 15 - 3)kte^{3t} + (12 - 5)ke^{3t} = 6e^{3t}$

$\Rightarrow 7ke^{3t} = 6e^{3t}$

$\Rightarrow 7k = 6$

$\Rightarrow k = \frac{6}{7}$

Particular Integral:

$x = \frac{6}{7}te^{3t}$

General Solution:

$x = Ae^{-\frac{1}{2}t} + Be^{3t} + \frac{6}{7}te^{3t}$

20) b)

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + y = 5\mathrm{e}^x$

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$

Auxiliary Equation:

$4m^2 - 4m + 1 = 0$

$\Rightarrow (2m-1)^2 = 0$

$\Rightarrow m = \frac{1}{2}$

Complementary Function:

$y = Ax\mathrm{e}^{\frac{1}{2}x} + B\mathrm{e}^{\frac{1}{2}x}$

Try this Particular Integral:

$y = k\mathrm{e}^x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = k\mathrm{e}^x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = k\mathrm{e}^x$

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + y = 5\mathrm{e}^x$

$\Rightarrow 4(k\mathrm{e}^x) - 4(k\mathrm{e}^x) + (k\mathrm{e}^x) = 5\mathrm{e}^x$

$\Rightarrow (4-4+1)k\mathrm{e}^x = 5\mathrm{e}^x$

$\Rightarrow k = 5$

Particular Integral:

$y = 5\mathrm{e}^x$

General Solution:

$y = Ax\mathrm{e}^{\frac{1}{2}x} + B\mathrm{e}^{\frac{1}{2}x} + 5\mathrm{e}^x$

20) c)

$4\ddot{x} + 81x = 8\mathrm{e}^{\frac{1}{2}t}$

Homogeneous Equation:

$4\ddot{x} + 81x = 0$

Auxiliary Equation:

$4m^2 + 81 = 0$

$\Rightarrow m^2 = -\frac{81}{4}$

$\Rightarrow m = \pm\frac{9}{2}\mathrm{i}$

Complementary Function:

$x = A\cos\frac{9}{2}t + B\sin\frac{9}{2}t$

Try this Particular Integral:

$x = k\mathrm{e}^{\frac{1}{2}t}$

$\dot{x} = \frac{1}{2}k\mathrm{e}^{\frac{1}{2}t}$

$\ddot{x} = \frac{1}{4}k\mathrm{e}^{\frac{1}{2}t}$

$4\ddot{x} + 81x = 8\mathrm{e}^{\frac{1}{2}t}$

$\Rightarrow 4\left(\frac{1}{4}k\mathrm{e}^{\frac{1}{2}t}\right) + 81\left(k\mathrm{e}^{\frac{1}{2}t}\right) = 8\mathrm{e}^{\frac{1}{2}t}$

$\Rightarrow (1 + 81)k\mathrm{e}^{\frac{1}{2}t} = 8\mathrm{e}^{\frac{1}{2}t}$

$\Rightarrow 82k\mathrm{e}^{\frac{1}{2}t} = 8\mathrm{e}^{\frac{1}{2}t}$

$\Rightarrow 82k = 8$

$\Rightarrow k = \frac{8}{82} = \frac{4}{41}$

Particular Integral:

$x = \frac{4}{41}\mathrm{e}^{\frac{1}{2}t}$

General Solution:

$x = A\cos\frac{9}{2}t + B\sin\frac{9}{2}t + \frac{4}{41}\mathrm{e}^{\frac{1}{2}t}$

20) d)

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 24\frac{\mathrm{d}y}{\mathrm{d}x} + 52y = 17\mathrm{e}^{-x}$

Homogeneous Equation:

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 24\frac{\mathrm{d}y}{\mathrm{d}x} + 52y = 0$

Auxiliary Equation:

$9m^2 - 24m + 52 = 0$

$\Rightarrow 9\left(m^2 - \frac{8}{3}m\right) + 52 = 0$

$\Rightarrow 9\left(\left(m - \frac{4}{3}\right)^2 - \frac{16}{9}\right) + 52 = 0$

$\Rightarrow 9\left(m - \frac{4}{3}\right)^2 - 16 + 52 = 0$

$\Rightarrow \left(m - \frac{4}{3}\right)^2 = -4$

$\Rightarrow m = \frac{4}{3} \pm 2\mathrm{i}$

Complementary Function:

$y = \mathrm{e}^{\frac{4}{3}x}(A\cos 2x + B\sin 2x)$

Try this Particular Integral:

$y = k\mathrm{e}^{-x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -k\mathrm{e}^{-x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = k\mathrm{e}^{-x}$

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 24\frac{\mathrm{d}y}{\mathrm{d}x} + 52y = 17\mathrm{e}^{-x}$

$\Rightarrow 9(k\mathrm{e}^{-x}) - 24(-k\mathrm{e}^{-x}) + 52(k\mathrm{e}^{-x}) = 17\mathrm{e}^{-x}$

$\Rightarrow (9 + 24 + 52)k\mathrm{e}^{-x} = 17\mathrm{e}^{-x}$

$\Rightarrow 85k = 17$

$\Rightarrow k = \frac{1}{5}$

Particular Integral:

$y = \frac{1}{5}\mathrm{e}^{-x}$

General Solution:

$y = \mathrm{e}^{\frac{4}{3}x}(A\cos 2x + B\sin 2x) + \frac{1}{5}\mathrm{e}^{-x}$

21)

$\ddot{x} + 9x = 9\mathrm{e}^{-2t}$

Homogeneous Equation:

$\ddot{x} + 9x = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3\mathrm{i}$

Complementary Function:

$x = A\cos 3t + B\sin 3t$

Try this Particular Integral:

$x = k\mathrm{e}^{-2t}$

$\dot{x} = -2k\mathrm{e}^{-2t}$

$\ddot{x} = 4k\mathrm{e}^{-2t}$

$\ddot{x} + 9x = 9\mathrm{e}^{-2t}$

$\Rightarrow (4k\mathrm{e}^{-2t}) + 9(k\mathrm{e}^{-2t}) = 9\mathrm{e}^{-2t}$

$\Rightarrow (4+9)k\mathrm{e}^{-2t} = 9\mathrm{e}^{-2t}$

$\Rightarrow 13k\mathrm{e}^{-2t} = 9\mathrm{e}^{-2t}$

$\Rightarrow 13k = 9$

$\Rightarrow k = \frac{9}{13}$

Particular Integral:

$x = \frac{9}{13}\mathrm{e}^{-2t}$

General Solution:

$x = A\cos 3t + B\sin 3t + \frac{9}{13}\mathrm{e}^{-2t}$

Substitute $x = 0$ and $t = 0$:

$0 = A\cos 0 + B\sin 0 + \frac{9}{13}\mathrm{e}^{0}$

$\Rightarrow 0 = A + \frac{9}{13} \Rightarrow A = -\frac{9}{13}$

$\therefore x = -\frac{9}{13}\cos 3t + B\sin 3t + \frac{9}{13}\mathrm{e}^{-2t}$

$\dot{x} = \frac{27}{13}\sin 3t + 3B\cos 3t - \frac{18}{13}\mathrm{e}^{-2t}$

Substitute $\dot{x} = 9$ and $t = 0$:

$9 = \frac{27}{13}\sin 0 + 3B\cos 0 - \frac{18}{13}\mathrm{e}^{0}$

$\Rightarrow 9 = 3B - \frac{18}{13} \Rightarrow B = \frac{45}{13}$

Particular Solution:

$x = -\frac{9}{13}\cos 3t + \frac{45}{13}\sin 3t + \frac{9}{13}e^{-2t}$

22)

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 60e^{-3x}$

Homogeneous Equation:

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 0$

Auxiliary Equation:

$m^2 + m - 12 = 0$

$\Rightarrow (m+4)(m-3) = 0$

$\Rightarrow m = -4$ or $m = 3$

Complementary Function:

$y = Ae^{-4x} + Be^{3x}$

Try this Particular Integral:

$y = ke^{-3x}$

$\frac{dy}{dx} = -3ke^{-3x}$

$\frac{d^2y}{dx^2} = 9ke^{-3x}$

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 60e^{-3x}$

$\Rightarrow (9ke^{-3x}) + (-3ke^{-3x}) - 12(ke^{-3x}) = 60e^{-3x}$

$\Rightarrow (9 - 3 - 12)ke^{-3x} = 60e^{-3x}$

$\Rightarrow -6ke^{-3x} = 60e^{-3x}$

$\Rightarrow -6k = 60$

$\Rightarrow k = -10$

Particular Integral:

$y = -10e^{-3x}$

General Solution:

$y = Ae^{-4x} + Be^{3x} - 10e^{-3x}$

As $x \to \infty$, $y \to 0$

This can only be true if $B = 0$, otherwise $e^{3x} \to \infty$ as $x \to \infty$

$\therefore y = Ae^{-4x} - 10e^{-3x}$

$\frac{dy}{dx} = -4Ae^{-4x} + 30e^{-3x}$

Substitute $\frac{dy}{dx} = 0$ and $x = 0$:

$0 = -4Ae^0 + 30e^0$

$\Rightarrow 0 = -4A + 30 \Rightarrow A = \frac{15}{2}$

Particular Solution:

$y = \frac{15}{2}e^{-4x} - 10e^{-3x}$

23) a)

Homogeneous Equation:

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$

Auxiliary Equation:

$m^2 - 4m + 4 = 0$

$\Rightarrow (m-2)^2 = 0$

$\Rightarrow m = 2$

Complementary Function:

$y = Axe^{2x} + Be^{2x}$

23) b)

$y = ke^{2x}$

$\frac{dy}{dx} = 2ke^{2x}$

$\frac{d^2y}{dx^2} = 4ke^{2x}$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 5e^{2x}$

$\Rightarrow 4ke^{2x} - 4(2ke^{2x}) + 4(ke^{2x}) = 5e^{2x}$

$\Rightarrow (4 - 8 + 4)ke^{2x} = 5e^{2x}$

$\Rightarrow 0 = 5$

There is no value of k for which this is true.

23) c)

$y = kxe^{2x}$

$\frac{dy}{dx} = 2kxe^{2x} + ke^{2x}$

$\frac{d^2y}{dx^2} = 4kxe^{2x} + 2ke^{2x} + 2ke^{2x}$

$= 4kxe^{2x} + 4ke^{2x}$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 5e^{2x}$

$\Rightarrow (4kxe^{2x} + 4ke^{2x}) - 4(2kxe^{2x} + ke^{2x}) + 4(kxe^{2x}) = 5e^{2x}$

$\Rightarrow (4 - 8 + 4)kxe^{2x} + (4 - 4)ke^{2x} = 5e^{2x}$

$\Rightarrow 0 + 0 = 5$

There is no value of k for which this is true.

23) d)

$y = kx^2e^{2x}$

$\frac{dy}{dx} = 2kx^2e^{2x} + 2kxe^{2x}$

$\frac{d^2y}{dx^2} = 4kx^2e^{2x} + 4kxe^{2x} + 4kxe^{2x} + 2ke^{2x}$

$= 4kx^2e^{2x} + 8kxe^{2x} + 2ke^{2x}$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 5e^{2x}$

$\Rightarrow (4kx^2e^{2x} + 8kxe^{2x} + 2ke^{2x}) - 4(2kx^2e^{2x} + 2kxe^{2x}) + 4(kx^2e^{2x}) = 5e^{2x}$

$\Rightarrow (4 - 8 + 4)kx^2e^{2x} + (8 - 8)kxe^{3x} + (2)ke^{3x} = 5e^{2x}$

$\Rightarrow 0 + 0 + 2ke^{2x} = 5e^{2x}$

$\Rightarrow 2k = 5$

$\Rightarrow k = \frac{5}{2}$

Particular Integral:

$y = \frac{5}{2}x^2e^{2x}$

$\therefore k = \frac{5}{2}$

23) e)

General Solution:

$y = Axe^{2x} + Be^{2x} + \frac{5}{2}x^2e^{2x}$

24)

$y = p\cos 3x + q\sin 3x$

$\frac{dy}{dx} = -3p\sin 3x + 3q\cos 3x$

$\frac{d^2y}{dx^2} = -9p\cos 3x - 9q\sin 3x$

$\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 9y = \sin 3x$

$\Rightarrow (-9p\cos 3x - 9q\sin 3x) + 10(-3p\sin 3x + 3q\cos 3x) + 9(p\cos 3x + q\sin 3x) = \sin 3x$

$\Rightarrow (-9p + 30q + 9p)\cos 3x + (-9q - 30p + 9p)\sin 3x = \sin 3x$

$\Rightarrow 30q\cos 3x - 30p\sin 3x = \sin 3x$

Compare coefficients:

$30q = 0 \Rightarrow q = 0$

$-30p = 1 \Rightarrow p = -\frac{1}{30}$

Particular Integral:

$y = -\frac{1}{30}\cos 3x$

25)

$y = p\cos 4x + q\sin 4x$

$\frac{dy}{dx} = -4p\sin 4x + 4q\cos 4x$

$\frac{d^2y}{dx^2} = -16p\cos 4x - 16q\sin 4x$

$\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 36y = 5\cos 4x$

$\Rightarrow (-16p\cos 4x - 16q\sin 4x) + 9(-4p\sin 4x + 4q\cos 4x) - 36(p\cos 4x + q\sin 4x) = 5\cos 4x$

$\Rightarrow (-16p + 36q - 36p)\cos 4x + (-16q - 36p - 36q)\sin 4x = 5\cos 4x$

$\Rightarrow (-52p + 36q)\cos 4x + (-36p - 52q)\sin 4x = 5\cos 4x$

Compare coefficients:

$-52p + 36q = 5$ (1)

$-36p - 52q = 0$ (2)

Solve (1) & (2):

$p = -\frac{13}{200}$ and $q = \frac{9}{200}$

Particular Integral:

$y = -\frac{13}{200}\cos 4x + \frac{9}{200}\sin 4x$

26) a)

$y = p\cos 6x + q\sin 6x$

$\frac{dy}{dx} = -6p\sin 6x + 6q\cos 6x$

$\frac{d^2y}{dx^2} = -36p\cos 6x - 36q\sin 6x$

$25\frac{d^2y}{dx^2} - 9y = 9\cos 6x$

$\Rightarrow 25(-36p\cos 6x - 36q\sin 6x) - 9(p\cos 6x + q\sin 6x) = 9\cos 6x$

$\Rightarrow (-900p - 9p)\cos 6x + (-900q - 9q)\sin 6x = 9\cos 6x$

$\Rightarrow -909p\cos 6x - 909q\sin 6x = 9\cos 6x$

Compare coefficients:

$-909p = 9 \Rightarrow p = -\frac{1}{101}$

$-909q = 0 \Rightarrow q = 0$

Particular Integral:

$y = -\frac{1}{101}\cos 6x$

26) b)

$y = p\cos 8x + q\sin 8x$

$\frac{dy}{dx} = -8p\sin 8x + 8q\cos 8x$

$\frac{d^2y}{dx^2} = -64p\cos 8x - 64q\sin 8x$

$2\frac{d^2y}{dx^2} - 11\frac{dy}{dx} + 5y = 89\sin 8x$

$\Rightarrow 2(-64p\cos 8x - 64q\sin 8x) - 11(-8p\sin 8x + 8q\cos 8x) + 5(p\cos 8x + q\sin 8x) = 89\sin 8x$

$\Rightarrow (-128p - 88q + 5p)\cos 8x + (-128q + 88p + 5q)\sin 8x = 89\sin 8x$

$\Rightarrow (-123p - 88q)\cos 8x + (88p - 123q)\sin 8x = 89\sin 8x$

Compare coefficients:

$-123p - 88q = 0$ (1)

$88p - 123q = 89$ (2)

Solve (1) & (2):

$p = \frac{88}{257}$ and $q = -\frac{123}{257}$

Particular Integral:

$y = \frac{88}{257}\cos 8x - \frac{123}{257}\sin 8x$

26) c)

$y = p\cos 2x + q\sin 2x$

$\frac{dy}{dx} = -2p\sin 2x + 2q\cos 2x$

$\frac{d^2y}{dx^2} = -4p\cos 2x - 4q\sin 2x$

$3\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 6y = 20\sin 2x - 4\cos 2x$

$\Rightarrow 3(-4p\cos 2x - 4q\sin 2x) + 11(-2p\sin 2x + 2q\cos 2x) + 6(p\cos 2x + q\sin 2x) = 20\sin 2x - 4\cos 2x$

$\Rightarrow (-12p + 22q + 6p)\cos 2x + (-12q - 22p + 6q)\sin 2x = 20\sin 2x - 4\cos 2x$

$\Rightarrow (-6p + 22q)\cos 2x + (-22p - 6q)\sin 2x = 20\sin 2x - 4\cos 2x$

Compare coefficients:

$-6p + 22q = -4$ (1)

$-22p - 6q = 20$ (2)

Solve (1) & (2):

$p = -\frac{4}{5}$ and $q = -\frac{2}{5}$

Particular Integral:

$y = -\frac{4}{5}\cos 2x - \frac{2}{5}\sin 2x$

27) a)

$25\frac{d^2y}{dx^2} - 9y = 9\cos 6x$

Homogeneous Equation:

$25\frac{d^2y}{dx^2} - 9y = 0$

Auxiliary Equation:

$25m^2 - 9 = 0$

$\Rightarrow (5m - 3)(5m + 3) = 0$

$\Rightarrow m = \frac{3}{5}$ or $m = -\frac{3}{5}$

Complementary Function:

$y = Ae^{\frac{3}{5}x} + Be^{-\frac{3}{5}x}$

Particular Integral:

$y = -\frac{1}{101}\cos 6x$

General Solution:

$y = Ae^{\frac{3}{5}x} + Be^{-\frac{3}{5}x} - \frac{1}{101}\cos 6x$

27) b)

$2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 11\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 89\sin 8x$

Homogeneous Equation:

$2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 11\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$2m^2 - 11m + 5 = 0$

$\Rightarrow (2m-1)(m-5) = 0$

$\Rightarrow m = \frac{1}{2}$ or $m = 5$

Complementary Function:

$y = A\mathrm{e}^{\frac{1}{2}x} + B\mathrm{e}^{5x}$

Particular Integral:

$y = \frac{88}{257}\cos 8x - \frac{123}{257}\sin 8x$

General Solution:

$y = A\mathrm{e}^{\frac{1}{2}x} + B\mathrm{e}^{5x} + \frac{88}{257}\cos 8x - \frac{123}{257}\sin 8x$

27) c)

$3\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 20\sin 2x - 4\cos 2x$

Homogeneous Equation:

$3\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$

Auxiliary Equation:

$3m^2 + 11m + 6 = 0$

$\Rightarrow (3m+2)(m+3) = 0$

$\Rightarrow m = -\frac{2}{3}$ or $m = -3$

Complementary Function:

$y = A\mathrm{e}^{-\frac{2}{3}x} + B\mathrm{e}^{-3x}$

Particular Integral:

$y = -\frac{4}{5}\cos 2x - \frac{2}{5}\sin 2x$

General Solution:

$y = A\mathrm{e}^{-\frac{2}{3}x} + B\mathrm{e}^{-3x} - \frac{4}{5}\cos 2x - \frac{2}{5}\sin 2x$

28) a)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

Auxiliary Equation:

$m^2 + 4 = 0$

$\Rightarrow m^2 = -4$

$\Rightarrow m = \pm 2\mathrm{i}$

Complementary Function:

$y = A\cos 2x + B\sin 2x$

28) b)

$y = p\cos 2x + q\sin 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2p\sin 2x + 2q\cos 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4p\cos 2x - 4q\sin 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = \sin 2x$

$\Rightarrow (-4p\cos 2x - 4q\sin 2x) + 4(p\cos 2x + q\sin 2x) = \sin 2x$

$\Rightarrow (-4p + 4p)\cos 2x + (-4q + 4q)\sin 2x = \sin 2x$

$\Rightarrow 0 + 0 = 1$

There is no value of k for which this is true.

28) c)

$y = px\cos 2x + qx\sin 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2px\sin 2x + p\cos 2x + 2qx\cos 2x + q\sin 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4px\cos 2x - 2p\sin 2x - 2p\sin 2x - 4qx\sin 2x + 2q\cos 2x + 2q\cos 2x$

$= -4px\cos 2x - 4p\sin 2x - 4qx\sin 2x + 4q\cos 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = \sin 2x$

$\Rightarrow (-4px\cos 2x - 4p\sin 2x - 4qx\sin 2x + 4q\cos 2x) + 4(px\cos 2x + qx\sin 2x) = \sin 2x$

$\Rightarrow (-4p + 4p)x\cos 2x + (4q)\cos 2x + (-4q + 4q)x\sin 2x + (-4p)\sin 2x = \sin 2x$

$\Rightarrow 4q\cos 2x - 4p\sin 2x = \sin 2x$

Compare coefficients:

$4q = 0 \Rightarrow q = 0$

$-4p = 1 \Rightarrow p = -\frac{1}{4}$

Particular Integral:

$y = -\frac{1}{4}x\cos 2x$

$\therefore p = -\frac{1}{4}$ and $q = 0$

28) d)

General Solution:

$y = A\cos 2x + B\sin 2x - \frac{1}{4}x\cos 2x$

29) a)

$y = k$

29) b)

$y = k\mathrm{e}^{3x}$

29) c)

$y = ax + b$

29) d)

$y = p\sin x + q\cos x$

29) e)

$y = ax^2 + bx + c$

29) f)

$y = kx\mathrm{e}^{-x}$

29) g)

$y = p\cos 4x + q\sin 4x$

29) h)

$y = ax + b$

29) i)

$y = p\cos 7x + q\sin 7x$

29) j)

$y = kx\mathrm{e}^{-2x}$

30) a)

$y = k\mathrm{e}^{-x}$

30) b)

$y = ax + b$

30) c)

$y = p\cos 2x + q\sin 2x$

30) d)

$y = ax^2 + bx + c$

30) e)

$y = k$

30) f)

$y = px\cos 6x + qx\sin 6x$

30) g)

$y = k\mathrm{e}^{6x}$

30) h)

$y = px\cos 6x + qx\sin 6x$

30) i)

$y = p\cos 9x + q\sin 9x$

30) j)

$y = px\cos 6x + qx\sin 6x$

31) a)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9\frac{\mathrm{d}y}{\mathrm{d}x} - 52y = 0$

Auxiliary Equation:

$m^2 + 9m - 52 = 0$

$\Rightarrow (m + 13)(m - 4) = 0$

$\Rightarrow m = -13$ or $m = 4$

Complementary Function:

$y = A\mathrm{e}^{-13x} + B\mathrm{e}^{4x}$

Try this Particular Integral:

$y = ax + b$

$\frac{\mathrm{d}y}{\mathrm{d}x} = a$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9\frac{\mathrm{d}y}{\mathrm{d}x} - 52y = 12 - 3x$

$\Rightarrow 0 + 9(a) - 52(ax + b) = 12 - 13x$

$\Rightarrow (-52a)x + (9a - 52b) = -13x + 12$

Compare coefficients:

$-52a = -13 \Rightarrow a = \frac{1}{4}$

$9\left(\frac{1}{4}\right) - 52b = 12 \Rightarrow b = -\frac{3}{16}$

Particular Integral:

$y = \frac{1}{4}x - \frac{3}{16}$

General Solution:

$y = A\mathrm{e}^{-13x} + B\mathrm{e}^{4x} + \frac{1}{4}x - \frac{3}{16}$

31) b)

Try this Particular Integral:

$y = ke^{5x}$

$\frac{dy}{dx} = 5ke^{5x}$

$\frac{d^2y}{dx^2} = 25ke^{5x}$

$\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 52y = 3e^{5x}$

$\Rightarrow (25ke^{5x}) + 9(5ke^{5x}) - 52(ke^{5x}) = 3e^{5x}$

$\Rightarrow (25 + 45 - 52)ke^{5x} = 3e^{5x}$

$\Rightarrow 18k = 3$

$\Rightarrow k = \frac{1}{6}$

Particular Integral:

$y = \frac{1}{6}e^{5x}$

General Solution:

$y = Ae^{-13x} + Be^{4x} + \frac{1}{6}e^{5x}$

31) c)

Try this Particular Integral:

$y = kxe^{4x}$

$\frac{dy}{dx} = 4kxe^{4x} + ke^{4x}$

$\frac{d^2y}{dx^2} = 16kxe^{4x} + 4ke^{4x} + 4ke^{4x}$

$= 16kxe^{4x} + 8ke^{4x}$

$\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 52y = 17e^{4x}$

$\Rightarrow (16kxe^{4x} + 8ke^{4x}) + 9(4kxe^{4x} + ke^{4x}) - 52(kxe^{4x}) = 17e^{4x}$

$\Rightarrow (16 + 36 - 52)kxe^{4x} + (8 + 9)ke^{4x} = 17e^{4x}$

$\Rightarrow 0 + 17ke^{4x} = 17e^{4x}$

$\Rightarrow 17k = 17$

$\Rightarrow k = 1$

Particular Integral:

$y = xe^{4x}$

General Solution:

$y = Ae^{-13x} + Be^{4x} + xe^{4x}$

31) d)

Try this Particular Integral:

$y = p\cos 3x + q\sin 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -3p\sin 3x + 3q\cos 3x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -9p\cos 3x - 9q\sin 3x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9\frac{\mathrm{d}y}{\mathrm{d}x} - 52y = 89\cos 3x + 89\sin 3x$

$\Rightarrow (-9p\cos 3x - 9q\sin 3x) + 9(-3p\sin 3x + 3q\cos 3x) - 52(p\cos 3x + q\sin 3x) = 89\cos 3x + 89\sin 3x$

$\Rightarrow (-9p + 27q - 52p)\cos 3x + (-9q - 27p - 52q)\sin 3x = 89\cos 3x + 89\sin 3x$

$\Rightarrow (-61p + 27q)\cos 3x + (-27p - 61q)\sin 3x = 89\cos 3x + 89\sin 3x$

Compare coefficients:

$-61p + 27q = 89$ (1)

$-27p - 61q = 89$ (2)

Solve (1) & (2):

$p = -\frac{44}{25}$ and $q = -\frac{17}{25}$

Particular Integral:

$y = -\frac{44}{25}\cos 3x - \frac{17}{25}\sin 3x$

General Solution:

$y = A\mathrm{e}^{-13x} + B\mathrm{e}^{4x} - \frac{44}{25}\cos 3x - \frac{17}{25}\sin 3x$

32) a)

Homogeneous Equation:

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 20\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 0$

Auxiliary Equation:

$4m^2 - 20m + 25 = 0$

$\Rightarrow (2m-5)^2 = 0$

$\Rightarrow m = \frac{5}{2}$

Complementary Function:

$y = Ax\mathrm{e}^{\frac{5}{2}x} + B\mathrm{e}^{\frac{5}{2}x}$

Try this Particular Integral:

$y = p\cos x + q\sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -p\sin x + q\cos x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -p\cos x - q\sin x$

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 20\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 841\cos x$

$\Rightarrow 4(-p\cos x - q\sin x) - 20(-p\sin x + q\cos x) + 25(p\cos x + q\sin x) = 841\cos x$

$\Rightarrow (-4p - 20q + 25p)\cos x + (-4q + 20p + 25q)\sin x = 841\cos x$

$\Rightarrow (21p - 20q)\cos x + (20p + 21q)\sin x = 841\cos x$

Compare coefficients:

$21p - 20q = 841$ (1)

$20p + 21q = 0$ (2)

Solve (1) & (2):

$p = 21$ and $q = -20$

Particular Integral:

$y = 21\cos x - 20\sin x$

General Solution:

$y = Ax\mathrm{e}^{\frac{5}{2}x} + B\mathrm{e}^{\frac{5}{2}x} + 21\cos x - 20\sin x$

32) b)

Try this Particular Integral:

$y = ke^x$

$\frac{dy}{dx} = ke^x$

$\frac{d^2y}{dx^2} = ke^x$

$4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = 18e^x$

$\Rightarrow 4(ke^x) - 20(ke^x) + 25(ke^x) = 18e^x$

$\Rightarrow (4 - 20 + 25)ke^x = 18e^x$

$\Rightarrow 9ke^x = 18e^x$

$\Rightarrow 9k = 18$

$\Rightarrow k = 2$

Particular Integral:

$y = 2e^x$

General Solution:

$y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} + 2e^x$

32) c)

Try this Particular Integral:

$y = ax + b$

$\frac{dy}{dx} = a$

$\frac{d^2y}{dx^2} = 0$

$4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = 25 - 125x$

$\Rightarrow 4(0) - 20(a) + 25(ax + b) = 25 - 125x$

$\Rightarrow (25a)x + (-20a + 25b) = -125x + 25$

Compare coefficients:

$25a = -125 \Rightarrow a = -5$

$-20(-5) + 25b = 25 \Rightarrow b = -3$

Particular Integral:

$y = -5x - 3$

General Solution:

$y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} - 5x - 3$

32) d)

Try this Particular Integral:

$y = kx^2 e^{\frac{5}{2}x}$

$\frac{dy}{dx} = \frac{5}{2}kx^2 e^{\frac{5}{2}x} + 2kxe^{\frac{5}{2}x}$

$\frac{d^2y}{dx^2} = \frac{25}{4}kx^2 e^{\frac{5}{2}x} + 5kxe^{\frac{5}{2}x} + 5kxe^{\frac{5}{2}x} + 2ke^{\frac{5}{2}x}$

$= \frac{25}{4}kx^2 e^{\frac{5}{2}x} + 10kxe^{\frac{5}{2}x} + 2ke^{\frac{5}{2}x}$

$4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = e^{\frac{5}{2}x}$

$\Rightarrow 4\left(\frac{25}{4}kx^2 e^{\frac{5}{2}x} + 10kxe^{\frac{5}{2}x} + 2ke^{\frac{5}{2}x}\right) - 20\left(\frac{5}{2}kx^2 e^{\frac{5}{2}x} + 2kxe^{\frac{5}{2}x}\right) + 25\left(kx^2 e^{\frac{5}{2}x}\right) = e^{\frac{5}{2}x}$

$\Rightarrow (25 - 50 + 25)kx^2 e^{\frac{5}{2}x} + (40 - 40)kxe^{\frac{5}{2}x} + (8)ke^{\frac{5}{2}x} = e^{\frac{5}{2}x}$

$\Rightarrow 0 + 0 + 8ke^{\frac{5}{2}x} = e^{\frac{5}{2}x}$

$\Rightarrow 8k = 1$

$\Rightarrow k = \frac{1}{8}$

Particular Integral:

$y = \frac{1}{8}x^2 e^{\frac{5}{2}x}$

General Solution:

$y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} + \frac{1}{8}x^2 e^{\frac{5}{2}x}$

33) a)

Homogeneous Equation:

$\ddot{x} + 2\dot{x} + 2x = 0$

Auxiliary Equation:

$m^2 + 2m + 2 = 0$

$\Rightarrow (m+1)^2 - 1 + 2 = 0$

$\Rightarrow (m+1)^2 = -1$

$\Rightarrow m = -1 \pm \mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t)$

Try this Particular Integral:

$x = k\mathrm{e}^{-t}$

$\dot{x} = -k\mathrm{e}^{-t}$

$\ddot{x} = k\mathrm{e}^{-t}$

$\ddot{x} + 2\dot{x} + 2x = 2\mathrm{e}^{-t}$

$\Rightarrow (k\mathrm{e}^{-t}) + 2(-k\mathrm{e}^{-t}) + 2(k\mathrm{e}^{-t}) = 2\mathrm{e}^{-t}$

$\Rightarrow (1 - 2 + 2)k\mathrm{e}^{-t} = 2\mathrm{e}^{-t}$

$\Rightarrow k\mathrm{e}^{-t} = 2\mathrm{e}^{-t}$

$\Rightarrow k = 2$

Particular Integral:

$x = 2\mathrm{e}^{-t}$

General Solution:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t) + 2\mathrm{e}^{-t}$

$\Rightarrow x = \mathrm{e}^{-t}(A\cos t + B\sin t + 2)$

Substitute $x = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(A\cos 0 + B\sin 0 + 2)$

$\Rightarrow 0 = A + 0 + 2 \Rightarrow A = -2$

$\therefore x = \mathrm{e}^{-t}(-2\cos t + B\sin t + 2)$

$\dot{x} = \mathrm{e}^{-t}(2\sin t + B\cos t) - \mathrm{e}^{-t}(-2\cos t + B\sin t + 2)$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(2\sin 0 + B\cos 0) - \mathrm{e}^{0}(-2\cos 0 + B\sin 0 + 2)$

$\Rightarrow 0 = 0 + B - (-2 + 0 + 2) \Rightarrow B = 0$

Particular Solution:

$x = \mathrm{e}^{-t}(-2\cos t + 2)$

$\Rightarrow x = -2\mathrm{e}^{-t}(\cos t - 1)$

33) b)

Try this Particular Integral:

$x = p\cos t + q\sin t$

$\dot{x} = -p\sin t + q\cos t$

$\ddot{x} = -p\cos t - q\sin t$

$\ddot{x} + 2\dot{x} + 2x = 5\sin t$

$\Rightarrow (-p\cos t - q\sin t) + 2(-p\sin t + q\cos t) + 2(p\cos t + q\sin t) = 5\sin t$

$\Rightarrow (-p + 2q + 2p)\cos t + (-q - 2p + 2q)\sin t = 5\sin t$

$\Rightarrow (p + 2q)\cos t + (-2p + q)\sin t = 5\sin t$

Compare coefficients:

$p + 2q = 0$ (1)

$-2p + q = 5$ (2)

Solve (1) & (2):

$p = -2$ and $q = 1$

Particular Integral:

$x = -2\cos t + \sin t$

General Solution:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t) - 2\cos t + \sin t$

Substitute $x = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) - 2\cos 0 + \sin 0$

$\Rightarrow 0 = A + 0 - 2 + 0 \Rightarrow A = 2$

$\therefore x = \mathrm{e}^{-t}(2\cos t + B\sin t) - 2\cos t + \sin t$

$\dot{x} = \mathrm{e}^{-t}(-2\sin t + B\cos t) - \mathrm{e}^{-t}(2\cos t + B\sin t) + 2\sin t + \cos t$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(-2\sin 0 + B\cos 0) - \mathrm{e}^{0}(2\cos 0 + B\sin 0) + 2\sin 0 + \cos 0$

$\Rightarrow 0 = 0 + B - (2 + 0) + 0 + 1 \Rightarrow B = 1$

Particular Solution:

$x = \mathrm{e}^{-t}(2\cos t + \sin t) - 2\cos t + \sin t$

33) c)

Try this Particular Integral:

$x = at^2 + bt + c$

$\dot{x} = 2at + b$

$\ddot{x} = 2a$

$\ddot{x} + 2\dot{x} + 2x = 6t^2$

$\Rightarrow (2a) + 2(2at + b) + 2(at^2 + bt + c) = 6t^2$

$\Rightarrow (2a)t^2 + (4a + 2b)t + (2a + 2b + 2c) = 6t^2$

Compare coefficients:

$2a = 6 \Rightarrow a = 3$

$4 \times 3 + 2b = 0 \Rightarrow b = -6$

$2 \times 3 + 2 \times -6 + 2c = 0 \Rightarrow c = 3$

Particular Integral:

$x = 3t^2 - 6t + 3$

General Solution:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t) + 3t^2 - 6t + 3$

Substitute $x = 0$ and $t = 0$:

$0 = \mathrm{e}^0(A\cos 0 + B\sin 0) + 3 \times 0^2 - 6 \times 0 + 3$

$\Rightarrow 0 = A + 0 + 0 - 0 + 3 \Rightarrow A = -3$

$\therefore x = \mathrm{e}^{-t}(-3\cos t + B\sin t) + 3t^2 - 6t + 3$

$\dot{x} = \mathrm{e}^{-t}(3\sin t + B\cos t) - \mathrm{e}^{-t}(-3\cos t + B\sin t) + 6t - 6$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \mathrm{e}^0(3\sin 0 + B\cos 0) - \mathrm{e}^0(-3\cos 0 + B\sin 0) + 6 \times 0 - 6$

$\Rightarrow 0 = 0 + B - (-3 + 0) + 0 - 6 \Rightarrow B = 3$

Particular Solution:

$x = \mathrm{e}^{-t}(-3\cos t + 3\sin t) + 3t^2 - 6t + 3$

33) d)

Try this Particular Integral:

$x = p\cosh t + q\sinh t$

$\dot{x} = p\sinh t + q\cosh t$

$\ddot{x} = p\cosh t + q\sinh t$

$\ddot{x} + 2\dot{x} + 2x = 4\sinh t$

$\Rightarrow (p\cosh t + q\sinh t) + 2(p\sinh t + q\cosh t) + 2(p\cosh t + q\sinh t) = 4\sinh t$

$\Rightarrow (p + 2q + 2p)\cosh t + (q + 2p + 2q)\sinh t = 4\sinh t$

$\Rightarrow (3p + 2q)\cosh t + (2p + 3q)\sinh t = 4\sinh t$

Compare coefficients:

$3p + 2q = 0$ (1)

$2p + 3q = 4$ (2)

Solve (1) & (2):

$p = -\frac{8}{5}$ and $q = \frac{12}{5}$

Particular Integral:

$x = -\frac{8}{5}\cosh t + \frac{12}{5}\sinh t$

General Solution:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t) - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t$

Substitute $x = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) - \frac{8}{5}\cosh 0 + \frac{12}{5}\sinh 0$

$\Rightarrow 0 = A + 0 - \frac{8}{5} + 0 \Rightarrow A = \frac{8}{5}$

$\therefore x = \mathrm{e}^{-t}\left(\frac{8}{5}\cos t + B\sin t\right) - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t$

$\dot{x} = \mathrm{e}^{-t}\left(-\frac{8}{5}\sin t + B\cos t\right) - \mathrm{e}^{-t}\left(\frac{8}{5}\cos t + B\sin t\right) - \frac{8}{5}\sinh t + \frac{12}{5}\cosh t$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}\left(-\frac{8}{5}\sin 0 + B\cos 0\right) - \mathrm{e}^{0}\left(\frac{8}{5}\cos 0 + B\sin 0\right) - \frac{8}{5}\sinh 0 + \frac{12}{5}\cosh 0$

$\Rightarrow 0 = 0 + B - \left(\frac{8}{5} + 0\right) - 0 + \frac{12}{5} \Rightarrow B = -\frac{4}{5}$

Particular Solution:

$x = \mathrm{e}^{-t}\left(\frac{8}{5}\cos t - \frac{4}{5}\sin t\right) - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t$

33) e)

Try this Particular Integral:

$x = p\cosh 2t + q\sinh 2t$

$\dot{x} = 2p\sinh 2t + 2q\cosh 2t$

$\ddot{x} = 4p\cosh 2t + 4q\sinh 2t$

$\ddot{x} + 2\dot{x} + 2x = 4\sinh t$

$\Rightarrow (4p\cosh 2t + 4q\sinh 2t) + 2(2p\sinh 2t + 2q\cosh 2t) + 2(p\cosh 2t + q\sinh 2t) = \cosh 2t - \sinh 2t$

$\Rightarrow (4p + 4q + 2p)\cosh 2t + (4q + 4p + 2q)\sinh 2t = \cosh 2t - \sinh 2t$

$\Rightarrow (6p + 4q)\cosh 2t + (4p + 6q)\sinh 2t = \cosh 2t - \sinh 2t$

Compare coefficients:

$6p + 4q = 1$ (1)

$4p + 6q = -1$ (2)

Solve (1) & (2):

$p = \frac{1}{2}$ and $q = -\frac{1}{2}$

Particular Integral:

$x = \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t$

General Solution:

$x = \mathrm{e}^{-t}(A\cos t + B\sin t) + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t$

Substitute $x = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + \frac{1}{2}\cosh 0 - \frac{1}{2}\sinh 0$

$\Rightarrow 0 = A + 0 + \frac{1}{2} - 0 \Rightarrow A = -\frac{1}{2}$

$\therefore x = \mathrm{e}^{-t}\left(-\frac{1}{2}\cos t + B\sin t\right) + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t$

$\dot{x} = \mathrm{e}^{-t}\left(\frac{1}{2}\sin t + B\cos t\right) - \mathrm{e}^{-t}\left(-\frac{1}{2}\cos t + B\sin t\right) + \sinh 2t - \cosh 2t$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}\left(\frac{1}{2}\sin 0 + B\cos 0\right) - \mathrm{e}^{0}\left(-\frac{1}{2}\cos 0 + B\sin 0\right) + \sinh 0 - \cosh 0$

$\Rightarrow 0 = 0 + B - \left(-\frac{1}{2} + 0\right) + 0 - 1 \Rightarrow B = \frac{1}{2}$

Particular Solution:

$x = \mathrm{e}^{-t}\left(-\frac{1}{2}\cos t + \frac{1}{2}\sin t\right) + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t$

34)

$y = pe^{2x} + qe^{4x}$

$\frac{dy}{dx} = 2pe^{2x} + 4qe^{4x}$

$\frac{d^2y}{dx^2} = 4pe^{2x} + 16qe^{4x}$

$5\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 2y = 4e^{2x} - 6e^{4x}$

$\Rightarrow 5(4pe^{2x} + 16qe^{4x}) + 9(2pe^{2x} + 4qe^{4x}) - 2(pe^{2x} + qe^{4x}) = 4e^{2x} - 6e^{4x}$

$\Rightarrow (20p + 18p - 2p)e^{2x} + (80q + 36q - 2q)e^{4x} = 4e^{2x} - 6e^{4x}$

$\Rightarrow 36pe^{2x} + 114qe^{4x} = 4e^{2x} - 6e^{4x}$

Compare coefficients:

$36p = 4 \Rightarrow p = \frac{1}{9}$

$114q = -6 \Rightarrow q = -\frac{1}{19}$

Particular Integral:

$y = \frac{1}{9}e^{2x} - \frac{1}{19}e^{4x}$

35)

$y = p + qx + r\cos 2x + k\sin 2x$

$\frac{dy}{dx} = q - 2r\sin 2x + 2k\cos 2x$

$\frac{d^2y}{dx^2} = -4r\cos 2x - 4k\sin 2x$

$4\frac{d^2y}{dx^2} + y = 9x + \cos 2x$

$\Rightarrow 4(-4r\cos 2x - 4k\sin 2x) + (p + qx + r\cos 2x + k\sin 2x) = 9x + \cos 2x$

$\Rightarrow (q)x + (p) + (-16r + r)\cos 2x + (-16k + k)\sin 2x = 9x + \cos 2x$

$\Rightarrow qx + p - 15r\cos 2x - 15k\sin 2x = 9x + \cos 2x$

Compare coefficients:

$q = 9$

$p = 0$

$-15r = 1 \Rightarrow r = -\frac{1}{15}$

$-15k = 0 \Rightarrow k = 0$

Particular Integral:

$y = 9x - \frac{1}{15}\cos 2x$

$\therefore p = 9,\ q = 0,\ r = -\frac{1}{15}$ and $k = 0$

36)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - y = 0$

Auxiliary Equation:

$m^2 - 1 = 0$

$\Rightarrow (m-1)(m+1) = 0$

$\Rightarrow m = 1$ or $m = -1$

Complementary Function:

$y = A\mathrm{e}^x + B\mathrm{e}^{-x}$

Try this Particular Integral:

$y = px + q + m\cos x + n\sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = p - m\sin x + n\cos x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -m\cos x - n\sin x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - y = x + \sin x$

$\Rightarrow (-m\cos x - n\sin x) - (px + q + m\cos x + n\sin x) = x + \sin x$

$\Rightarrow (-p)x + (-q) + (-m-m)\cos x + (-n-n)\sin x = x + \sin x$

$\Rightarrow (-p)x + (-q) + (-2m)\cos x + (-2n)\sin x = x + \sin x$

Compare coefficients:

$-p = 1 \Rightarrow p = -1$

$-q = 0 \Rightarrow q = 0$

$-2m = 0 \Rightarrow m = 0$

$-2n = 1 \Rightarrow n = -\frac{1}{2}$

Particular Integral:

$y = -x - \frac{1}{2}\sin x$

General Solution:

$y = A\mathrm{e}^x + B\mathrm{e}^{-x} - x - \frac{1}{2}\sin x$

37) a)

$y = p + q\mathrm{e}^x$

37) b)

$y = p\mathrm{e}^{2x} + q\mathrm{e}^x$

37) c)

$y = px + q + re^{5x}$

37) d)

$y = pe^{4x} + q\cos x + r\sin x$

37) e)

$y = p\cos 3x + q\sin 3x + rx^2 + sx + t$

37) f)

$y = p + qx + re^{-5x}$

37) g)

$y = p + qx + r\cos\frac{1}{2}x + k\sin\frac{1}{2}x$

37) h)

$y = pxe^{-6x} + q$

37) i)

$y = pe^{6x} + qxe^{7x} + rx + s$

37) j)

$y = pxe^{-6x} + qxe^{7x} + r\cos 3x + k\sin 3x$

38) a)

$y = p\cos 3x + q\sin 3x + r$

38) b)

$y = px + q + re^{-8x}$

38) c)

$y = pe^{x} + q\cos\frac{2}{3}x + r\sin\frac{2}{3}x$

38) d)

$y = pe^{4x} + qx + r$

38) e)

$y = p\cos 12x + q\sin 12x + r$

38) f)

$y = pe^{3x} + qe^{-2x} + rx^2 + mx + n$

38) g)

$y = p\cos x + q\sin x + m\cos 2x + n\sin 2x$

38) h)

$y = px^2 + qx + r + ke^{10x}$

38) i)

$y = px^2e^{8x} + qx^2 + rx + k$

38) j)

$y = p\cos 9x + q\sin 9x + m\cos 4x + n\sin 4x + ax^2 + bx + c + kx^2e^{8x}$

39)

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 0$

Auxiliary Equation:

$m^2 - 5m + 6 = 0$

$\Rightarrow (m-2)(m-3) = 0$

$\Rightarrow m = 2$ or $m = 3$

Complementary Function:

$y = A\mathrm{e}^{2x} + B\mathrm{e}^{3x}$

Try this Particular Integral:

$y = px\mathrm{e}^{2x} + qx\mathrm{e}^{3x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2px\mathrm{e}^{2x} + p\mathrm{e}^{2x} + 3qx\mathrm{e}^{3x} + q\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4px\mathrm{e}^{2x} + 2p\mathrm{e}^{2x} + 2p\mathrm{e}^{2x} + 9qx\mathrm{e}^{3x} + 3q\mathrm{e}^{3x} + 3q\mathrm{e}^{3x}$

$= 4px\mathrm{e}^{2x} + 4p\mathrm{e}^{2x} + 9qx\mathrm{e}^{3x} + 6q\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 3\mathrm{e}^{2x} + 2\mathrm{e}^{3x}$

$\Rightarrow (4px\mathrm{e}^{2x} + 4p\mathrm{e}^{2x} + 9qx\mathrm{e}^{3x} + 6q\mathrm{e}^{3x}) - 5(2px\mathrm{e}^{2x} + p\mathrm{e}^{2x} + 3qx\mathrm{e}^{3x} + q\mathrm{e}^{3x}) + 6(px\mathrm{e}^{2x} + qx\mathrm{e}^{3x}) = 3\mathrm{e}^{2x} + 2\mathrm{e}^{3x}$

$\Rightarrow (4p - 10p + 6p)x\mathrm{e}^{2x} + (4p - 5p)\mathrm{e}^{2x} + (9q - 15q + 6q)x\mathrm{e}^{3x} + (6q - 5q)\mathrm{e}^{3x} = 3\mathrm{e}^{2x} + 2\mathrm{e}^{3x}$

$\Rightarrow -p\mathrm{e}^{2x} + q\mathrm{e}^{3x} = 3\mathrm{e}^{2x} + 2\mathrm{e}^{3x}$

Compare coefficients:

$-p = 3 \Rightarrow p = -3$

$q = 2$

Particular Integral:

$y = -3x\mathrm{e}^{2x} + 2x\mathrm{e}^{3x}$

General Solution:

$y = A\mathrm{e}^{2x} + B\mathrm{e}^{3x} - 3x\mathrm{e}^{2x} + 2x\mathrm{e}^{3x}$

40)

Homogeneous Equation:

$25\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 40\frac{\mathrm{d}y}{\mathrm{d}x} + 16y = 0$

Auxiliary Equation:

$25m^2 - 40m + 16 = 0$

$\Rightarrow (5m-4)^2 = 0$

$\Rightarrow m = \frac{4}{5}$

Complementary Function:

$y = Ax\mathrm{e}^{\frac{4}{5}x} + B\mathrm{e}^{\frac{4}{5}x}$

Try this Particular Integral:

$y = k + p\mathrm{e}^{-\frac{2}{5}x} + qx^2\mathrm{e}^{\frac{4}{5}x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2}{5}p\mathrm{e}^{-\frac{2}{5}x} + \frac{4}{5}qx^2\mathrm{e}^{\frac{4}{5}x} + 2qx\mathrm{e}^{\frac{4}{5}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{4}{25}p\mathrm{e}^{-\frac{2}{5}x} + \frac{16}{25}qx^2\mathrm{e}^{\frac{4}{5}x} + \frac{8}{5}qx\mathrm{e}^{\frac{4}{5}x} + \frac{8}{5}qx\mathrm{e}^{\frac{4}{5}x} + 2q\mathrm{e}^{\frac{4}{5}x}$

$= \frac{4}{25}p\mathrm{e}^{-\frac{2}{5}x} + \frac{16}{25}qx^2\mathrm{e}^{\frac{4}{5}x} + \frac{16}{5}qx\mathrm{e}^{\frac{4}{5}x} + 2q\mathrm{e}^{\frac{4}{5}x}$

$25\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 40\frac{\mathrm{d}y}{\mathrm{d}x} + 16y = 4 - 9\mathrm{e}^{-\frac{2}{5}x} - 10\mathrm{e}^{\frac{4}{5}x}$

$\Rightarrow 25\left(\frac{4}{25}p\mathrm{e}^{-\frac{2}{5}x} + \frac{16}{25}qx^2\mathrm{e}^{\frac{4}{5}x} + \frac{16}{5}qx\mathrm{e}^{\frac{4}{5}x} + 2q\mathrm{e}^{\frac{4}{5}x}\right) - 40\left(-\frac{2}{5}p\mathrm{e}^{-\frac{2}{5}x} + \frac{4}{5}qx^2\mathrm{e}^{\frac{4}{5}x} + 2qx\mathrm{e}^{\frac{4}{5}x}\right) +$

$16\left(k + p\mathrm{e}^{-\frac{2}{5}x} + qx^2\mathrm{e}^{\frac{4}{5}x}\right) = 4 - 9\mathrm{e}^{-\frac{2}{5}x} - 10\mathrm{e}^{\frac{4}{5}x}$

$\Rightarrow (16k) + (4p + 16p + 16p)\mathrm{e}^{-\frac{2}{5}x} + (16q - 32q + 16q)x^2\mathrm{e}^{\frac{4}{5}x} + (80q - 80q)x\mathrm{e}^{\frac{4}{5}x} +$

$(50q)\mathrm{e}^{\frac{4}{5}x} = 4 - 9\mathrm{e}^{-\frac{2}{5}x} - 10\mathrm{e}^{\frac{4}{5}x}$

$\Rightarrow (16k) + (36p)\mathrm{e}^{-\frac{2}{5}x} + (50q)\mathrm{e}^{\frac{4}{5}x} = 4 - 9\mathrm{e}^{-\frac{2}{5}x} - 10\mathrm{e}^{\frac{4}{5}x}$

Compare coefficients:

$16k = 4 \Rightarrow k = \frac{1}{4}$

$36p = -9 \Rightarrow p = -\frac{1}{4}$

$50q = -10 \Rightarrow q = -\frac{1}{5}$

Particular Integral:

$y = \frac{1}{4} - \frac{1}{4}\mathrm{e}^{-\frac{2}{5}x} - \frac{1}{5}x^2\mathrm{e}^{\frac{4}{5}x}$

General Solution:

$y = Ax\mathrm{e}^{\frac{4}{5}x} + B\mathrm{e}^{\frac{4}{5}x} + \frac{1}{4} - \frac{1}{4}\mathrm{e}^{-\frac{2}{5}x} - \frac{1}{5}x^2\mathrm{e}^{\frac{4}{5}x}$

41) a)

Homogeneous Equation:

$\ddot{x} - 7\dot{x} + 6x = 0$

Auxiliary Equation:

$m^2 - 7m + 6 = 0$

$\Rightarrow (m-1)(m-6) = 0$

$\Rightarrow m = 1$ or $m = 6$

Complementary Function:

$x = Ae^t + Be^{6t}$

Try this Particular Integral:

$x = pt + q + r\cos t + k\sin t$

$\dot{x} = p - r\sin t + k\cos t$

$\ddot{x} = -r\cos t - k\sin t$

$\ddot{x} - 7\dot{x} + 6x = 72t + 74\sin t$

$\Rightarrow (-r\cos t - k\sin t) - 7(p - r\sin t + k\cos t) + 6(pt + q + r\cos t + k\sin t) = 72t + 74\sin t$

$\Rightarrow (6p)t + (-7p + 6q) + (-r - 7k + 6r)\cos t + (-k + 7r + 6k)\sin t = 72t + 74\sin t$

$\Rightarrow (6p)t + (-7p + 6q) + (5r - 7k)\cos t + (7r + 5k)\sin t = 72t + 74\sin t$

Compare coefficients:

$6p = 72 \Rightarrow p = 12$

$-7 \times 12 + 6q = 0 \Rightarrow q = 14$

$5r - 7k = 0$ (1)

$7r + 5k = 74$ (2)

Solve (1) & (2):

$r = 7$ and $k = 5$

Particular Integral:

$x = 12t + 14 + 7\cos t + 5\sin t$

General Solution:

$x = Ae^t + Be^{6t} + 12t + 14 + 7\cos t + 5\sin t$

Substitute $x = 0$ and $t = 0$:

$0 = Ae^0 + Be^0 + 12 \times 0 + 14 + 7\cos 0 + 5\sin 0$

$\Rightarrow 0 = A + B + 0 + 14 + 7 + 0$

$\Rightarrow A + B = -21$ (3)

$\dot{x} = Ae^t + 6Be^{6t} + 12 - 7\sin t + 5\cos t$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = Ae^0 + 6Be^0 + 12 - 7\sin 0 + 5\cos 0$

$\Rightarrow 1 = A + 6B + 12 - 0 + 5$

$\Rightarrow A + 6B = -16$ (4)

Solve (3) & (4):

$A = -22$ and $B = 1$

Particular Solution:

$x = -22e^t + e^{6t} + 12t + 14 + 7\cos t + 5\sin t$

41) b)

Try this Particular Integral:

$x = pte^t + qe^{2t}$

$\dot{x} = pte^t + pe^t + 2qe^{2t}$

$\ddot{x} = pte^t + pe^t + pe^t + 4qe^{2t}$

$= pte^t + 2pe^t + 4qe^{2t}$

$\ddot{x} - 7\dot{x} + 6x = 50e^t - 52e^{2t}$

$\Rightarrow (pte^t + 2pe^t + 4qe^{2t}) - 7(pte^t + pe^t + 2qe^{2t}) + 6(pte^t + qe^{2t}) = 50e^t - 52e^{2t}$

$\Rightarrow (p - 7p + 6p)te^t + (2p - 7p)e^t + (4q - 14q + 6q)e^{2t} = 50e^t - 52e^{2t}$

$\Rightarrow -5pe^t - 4qe^{2t} = 50e^t - 52e^{2t}$

Compare coefficients:

$-5p = 50 \Rightarrow p = -10$

$-4q = -52 \Rightarrow q = 13$

Particular Integral:

$x = -10te^t + 13e^{2t}$

General Solution:

$x = Ae^t + Be^{6t} - 10te^t + 13e^{2t}$

Substitute $x = 0$ and $t = 0$:

$0 = Ae^0 + Be^0 - 10 \times 0 \times e^0 + 13e^0$

$\Rightarrow 0 = A + B - 0 + 13$

$\Rightarrow A + B = -13$ (1)

$\dot{x} = Ae^t + 6Be^{6t} - 10te^t - 10e^t + 26e^{2t}$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = Ae^0 + 6Be^0 - 10 \times 0 \times e^0 - 10e^0 + 26e^0$

$\Rightarrow 1 = A + 6B - 0 - 10 + 26$

$\Rightarrow A + 6B = -15$ (2)

Solve (1) & (2):

$A = -\frac{63}{5}$ and $B = -\frac{2}{5}$

Particular Solution:

$x = -\frac{63}{5}e^t - \frac{2}{5}e^{6t} - 10te^t + 13e^{2t}$

41) c)

Try this Particular Integral:

$x = pte^{6t} + qt + r$

$\dot{x} = 6pte^{6t} + pe^{6t} + q$

$\ddot{x} = 36pte^{6t} + 6pe^{6t} + 6pe^{6t}$

$= 36pte^{6t} + 12pe^{6t}$

$\ddot{x} - 7\dot{x} + 6x = 35e^{6t} - 36t$

$\Rightarrow (36pte^{6t} + 12pe^{6t}) - 7(6pte^{6t} + pe^{6t} + q) + 6(pte^{6t} + qt + r) = 35e^{6t} - 36t$

$\Rightarrow (36p - 42p + 6p)te^{6t} + (12p - 7p)e^{6t} + (6q)t + (-7q + 6r) = 35e^{6t} - 36t$

$\Rightarrow 5pe^{6t} + 6qt + (-7q + 6r) = 35e^{6t} - 36t$

Compare coefficients:

$5p = 35 \Rightarrow p = 7$

$6q = -36 \Rightarrow q = -6$

$-7 \times -6 + 6r = 0 \Rightarrow r = -7$

Particular Integral:

$x = 7te^{6t} - 6t - 7$

General Solution:

$x = Ae^{t} + Be^{6t} + 7te^{6t} - 6t - 7$

Substitute $x = 0$ and $t = 0$:

$0 = Ae^{0} + Be^{0} + 7 \times 0 \times e^{0} - 6 \times 0 - 7$

$\Rightarrow 0 = A + B + 0 - 0 - 7$

$\Rightarrow A + B = 7$ (1)

$\dot{x} = Ae^{t} + 6Be^{6t} + 42te^{6t} + 7e^{6t} - 6$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = Ae^{0} + 6Be^{0} + 42 \times 0 \times e^{0} + 7e^{0} - 6$

$\Rightarrow 1 = A + 6B + 0 + 7 - 6$

$\Rightarrow A + 6B = 0$ (2)

Solve (1) & (2):

$A = \frac{42}{5}$ and $B = -\frac{7}{5}$

Particular Solution:

$x = \frac{42}{5}e^{t} - \frac{7}{5}e^{6t} + 7te^{6t} - 6t - 7$

41) d)

Try this Particular Integral:

$x = pt^2 + qt + r + m\cos 2t + n\sin 2t$

$\dot{x} = 2pt + q - 2m\sin 2t + 2n\cos 2t$

$\ddot{x} = 2p - 4m\cos 2t - 4n\sin 2t$

$\ddot{x} - 7\dot{x} + 6x = 108t^2 - 100\cos 2t$

$\Rightarrow (2p - 4m\cos 2t - 4n\sin 2t) - 7(2pt + q - 2m\sin 2t + 2n\cos 2t) +$
$6(pt^2 + qt + r + m\cos 2t + n\sin 2t) = 108t^2 - 100\cos 2t$

$\Rightarrow (6p)t^2 + (-14p + 6q)t + (2p - 7q + 6r) + (-4m - 14n + 6m)\cos 2t +$
$(-4n + 14m + 6n)\sin 2t = 108t^2 - 100\cos 2t$

$\Rightarrow (6p)t^2 + (-14p + 6q)t + (2p - 7q + 6r) + (2m - 14n)\cos 2t + (14m +$
$2n)\sin 2t = 108t^2 - 100\cos 2t$

Compare coefficients:

$6p = 108 \Rightarrow p = 18$

$-14 \times 18 + 6q = 0 \Rightarrow q = 42$

$2 \times 18 - 7 \times 42 + 6r = 0 \Rightarrow r = 43$

$2m - 14n = -100$ (1)

$14m + 2n = 0$ (2)

Solve (1) & (2)

$m = -1$ and $n = 7$

Particular Integral:

$x = 18t^2 + 42t + 43 - \cos 2t + 7\sin 2t$

General Solution:

$x = Ae^t + Be^{6t} + 18t^2 + 42t + 43 - \cos 2t + 7\sin 2t$

Substitute $x = 0$ and $t = 0$:

$0 = Ae^0 + Be^0 + 18 \times 0^2 + 42 \times 0 + 43 - \cos 0 + 7\sin 0$

$\Rightarrow 0 = A + B + 0 + 0 + 43 - 1 + 0$

$\Rightarrow A + B = -42$ (1)

$\dot{x} = Ae^t + 6Be^{6t} + 36t + 42 + 2\sin 2t + 14\cos 2t$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = Ae^0 + 6Be^0 + 36 \times 0 + 42 + 2\sin 0 + 14\cos 0$

$\Rightarrow 1 = A + 6B + 0 + 42 + 0 + 14$

$\Rightarrow A + 6B = -55$ (2)

Solve (1) & (2):

$A = -\frac{197}{5}$ and $B = -\frac{13}{5}$

Particular Solution:

$x = -\frac{197}{5}\mathrm{e}^{t} - \frac{13}{5}\mathrm{e}^{6t} + 18t^2 + 42t + 43 - \cos 2t + 7\sin 2t$

42)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = \sin x + \sin 2x$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = 0$

Auxiliary Equation:

$m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm\mathrm{i}$

Complementary Function:

$y = A\cos x + B\sin x$

Try this Particular Integral:

$y = px\cos x + qx\sin x + m\cos 2x + n\sin 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -px\sin x + p\cos x + qx\cos x + q\sin x - 2m\sin 2x + 2n\cos 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -px\cos x - p\sin x - p\sin x - qx\sin x + q\cos x + q\cos x - 4m\cos 2x -$
$4n\sin 2x$

$= -px\cos x - 2p\sin x - qx\sin x + 2q\cos x - 4m\cos 2x - 4n\sin 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = \sin x + \sin 2x$

$\Rightarrow (-px\cos x - 2p\sin x - qx\sin x + 2q\cos x - 4m\cos 2x - 4n\sin 2x) +$
$(px\cos x + qx\sin x + m\cos 2x + n\sin 2x) = \sin x + \sin 2x$

$\Rightarrow (-p + p)x\cos x + (-q + q)x\sin x + (2q)\cos x + (-2p)\sin x + (-4m + m)\cos 2x +$
$(-4n + n)\sin 2x = \sin x + \sin 2x$

$\Rightarrow (2q)\cos x + (-2p)\sin x + (-3m)\cos 2x + (-3n)\sin 2x = \sin x + \sin 2x$

Compare coefficients:

$2q = 0 \Rightarrow q = 0$

$-2p = 1 \Rightarrow p = -\frac{1}{2}$

$-3m = 0 \Rightarrow m = 0$

$-3n = 1 \Rightarrow n = -\frac{1}{3}$

Particular Integral:

$y = -\frac{1}{2}x\cos x - \frac{1}{3}\sin 2x$

General Solution:

$y = A\cos x + B\sin x - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x$

Substitute $y = 5$ and $x = 0$:

$5 = A\cos 0 + B\sin 0 - \frac{1}{2} \times 0 \times \cos 0 - \frac{1}{3}\sin 0$

$\Rightarrow 5 = A + 0 - 0 - 0 \Rightarrow A = 5$

$\therefore y = 5\cos x + B\sin x - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -5\sin x + B\cos x + \frac{1}{2}x\sin x - \frac{1}{2}\cos x - \frac{2}{3}\cos 2x$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$ and $x = 0$:

$3 = -5\sin 0 + B\cos 0 + \frac{1}{2} \times 0 \times \sin 0 - \frac{1}{2}\cos 0 - \frac{2}{3}\cos 0$

$\Rightarrow 3 = 0 + B + 0 - \frac{1}{2} - \frac{2}{3} \Rightarrow B = \frac{25}{6}$

Particular Solution:

$y = 5\cos x + \frac{25}{6}\sin x - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x$

43)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 17\cos 2x + 10\sin x$

Homogeneous Equation:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

Auxiliary Equation:

$m^2 + 2m + 5 = 0$

$\Rightarrow (m+1)^2 - 1 + 5 = 0$

$\Rightarrow (m+1)^2 = -4$

$\Rightarrow m = -1 \pm 2\mathrm{i}$

Complementary Function:

$y = \mathrm{e}^{-x}(A\cos 2x + B\sin 2x)$

Try this Particular Integral:

$y = p\cos 2x + q\sin 2x + m\cos x + n\sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2p\sin 2x + 2q\cos 2x - m\sin x + n\cos x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4p\cos 2x - 4q\sin 2x - m\cos x - n\sin x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 17\cos 2x + 10\sin x$

$\Rightarrow (-4p\cos 2x - 4q\sin 2x - m\cos x - n\sin x) + 2(-2p\sin 2x + 2q\cos 2x - m\sin x + n\cos x) + 5(p\cos 2x + q\sin 2x + m\cos x + n\sin x) = 17\cos 2x + 10\sin x$

$\Rightarrow (-4p + 4q + 5p)\cos 2x + (-4q - 4p + 5q)\sin 2x + (-m + 2n + 5m)\cos x + (-n - 2m + 5n)\sin x = 17\cos 2x + 10\sin x$

$\Rightarrow (p + 4q)\cos 2x + (-4p + q)\sin 2x + (4m + 2n)\cos x + (-2m + 4n)\sin x = 17\cos 2x + 10\sin x$

Compare coefficients:

$p + 4q = 17$ (1)

$-4p + q = 0$ (2)

Solve (1) & (2):

$p = 1$ and $q = 4$

$4m + 2n = 0$ (3)

$-2m + 4n = 10$ (4)

Solve (3) & (4):

$m = -1$ and $n = 2$

Particular Integral:

$y = \cos 2x + 4\sin 2x - \cos x + 2\sin x$

General Solution:

$y = \mathrm{e}^{-x}(A\cos 2x + B\sin 2x) + \cos 2x + 4\sin 2x - \cos x + 2\sin x$

Substitute $y = 10$ and $x = 0$:

$10 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + \cos 0 + 4\sin 0 - \cos 0 + 2\sin 0$

$\Rightarrow 10 = A + 0 + 1 + 0 - 1 + 0 \Rightarrow A = 10$

$\therefore y = \mathrm{e}^{-x}(10\cos 2x + B\sin 2x) + \cos 2x + 4\sin 2x - \cos x + 2\sin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-x}(-20\sin 2x + 2B\cos 2x) - \mathrm{e}^{-x}(10\cos 2x + B\sin 2x) - 2\sin 2x + 8\cos 2x + \sin x + 2\cos x$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and $x = 0$:

$0 = \mathrm{e}^{0}(-20\sin 0 + 2B\cos 0) - \mathrm{e}^{0}(10\cos 0 + B\sin 0) - 2\sin 0 + 8\cos 0 + \sin 0 + 2\cos 0$

$\Rightarrow 0 = 0 + 2B - 10 - 0 - 0 + 8 + 0 + 2 \Rightarrow B = 0$

Particular Solution:

$y = 10\mathrm{e}^{-x}\cos 2x + \cos 2x + 4\sin 2x - \cos x + 2\sin x$